

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report evaluates patent evidence relevant to low-acrylonitrile nitrile butadiene rubber (NBR), defined here as acrylonitrile-butadiene copolymers with reduced acrylonitrile incorporation relative to conventional grades. The required patent manifest contains one primary patent, US11896953B2, which does not disclose NBR elastomer synthesis but instead describes hydrophilic-lipophilic balance (HLB) copolymers for solid phase extraction. It is nevertheless compositionally relevant because it uses acrylonitrile as the hydrophilic monomer in a divinylbenzene-acrylonitrile copolymer and explicitly discusses monomer-ratio control, porogen selection, and reproducibility across a broad range of hydrophobic/hydrophilic feed ratios. The disclosed polymerization approach is a copolymerization of at least one hydrophobic monomer, preferably divinylbenzene, with at least one hydrophilic monomer, preferably acrylonitrile, in the presence of both nonpolar and polar porogens. The patent emphasizes that acrylonitrile is more effectively incorporated across a broader range of monomer ratios than N-vinylpyrrolidone, and that the resulting copolymers can be used in conventional or simplified SPE workflows without conditioning and equilibration steps. Key formulation trends include use of mixed porogens, especially o-xylene with 4-methyl-2-pentanol, and broad hydrophobic-to-hydrophilic monomer ratios from 10:90 to 90:10 wt.%, with a preferred 30:70 to 70:30 wt.% window. Although not a direct low-ACN NBR synthesis disclosure, the patent provides useful evidence on acrylonitrile incorporation control in copolymer systems and on how monomer ratio and porogen environment influence polymer structure and performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11896953B2
- Patent Title: HLB copolymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of hydrophilic-lipophilic balance copolymers by copolymerizing at least one hydrophobic monomer, preferably divinylbenzene, with at least one hydrophilic monomer, preferably acrylonitrile. The copolymerization is carried out in the presence of both nonpolar and polar porogens to tune pore structure and adsorbent performance. - Hydrophobic monomer: divinylbenzene — Abstract/Claims - Hydrophilic monomer: acrylonitrile — Abstract/Claims - Copolymerization medium: nonpolar porogen and polar porogen — Claim 1 - Nonpolar porogen examples: toluene, xylenes, benzene, hexane, cyclohexane, pentane, heptane, octane, nonane, decane, dodecane, isooctane, dichloromethane, chloroform, carbon tetrachloride — Claim 2 - Polar porogen examples: benzyl alcohol, butanol, pentanol, hexanol, heptanol, 4-methyl-2-pentanol, isoamyl alcohol, dodecanol, ethyl acetate, 2-ethylhexanol, cyclohexanol — Claim 3 - Preferred porogen pair: o-xylene and 4-methyl-2-pentanol — Claim 4 - Hydrophobic monomer to hydrophilic monomer ratio: 10:90 to 90:10 wt.% — Claim 5 - Preferred hydrophobic monomer to hydrophilic monomer ratio: 30:70 to 70:30 wt.% — Claim 6

Relevant Experimental Evidence - Example 1: HLB-01 is presented as an acrylonitrile-containing HLB copolymer and compared with a conventional vinylpyrrolidone-containing HLB material. - Example 2: HLB-44, HLB-48, and HLB-49 are shown with varying copolymer ratios to illustrate composition effects on pore structure. - Example 3: HLB-65 on silica particles is evaluated before and after washing with acetonitrile, demonstrating the silica coating process. - Example 4: HLB-67 is compared with conventional HLB products in SPE recovery studies using both 5-step and simplified 3-step protocols. - Example 5: HLB-79 coated on silica is compared with a conventional HLB adsorbent using a 3-step SPE protocol.

Technical Relevance This patent is relevant because it explicitly uses acrylonitrile as a copolymer monomer and provides concrete guidance on monomer-ratio control and porogen selection, both of which are central to controlling acrylonitrile incorporation in low-ACN copolymer systems. However, it is not a direct disclosure of NBR elastomer synthesis and should be treated as an indirect composition/process reference rather than a primary low-acrylonitrile NBR patent.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The available patent evidence does not disclose a direct low-acrylonitrile NBR elastomer synthesis route. The single primary patent instead demonstrates acrylonitrile-containing copolymer formation

in an HLB adsorbent context, with explicit control over monomer ratio and porogen environment. For low-ACN NBR development, the most transferable lesson is that acrylonitrile incorporation can be tuned through feed composition and solvent/porogen selection; however, no elastomer-specific polymerization conditions, initiators, or conversion data are provided here. Based on the evidence, the most defensible recommendation is to use controlled copolymer feed ratios and a mixed solvent environment when targeting reduced acrylonitrile incorporation, while recognizing that additional NBR-specific patents would be needed to define practical elastomer synthesis parameters.

5. REFERENCES

- US11896953B2 | HLB copolymers | Not disclosed | US | 2024 | <https://patents.google.com/patent/US11896953B2/en>